

Worksheet · Section 5.3–5.4

Computing Bias · Crowdsourcing



The Big Picture

Algorithms can repeat human bias, and crowdsourcing can gather huge amounts of data — if the crowd is representative. Practice finding where bias comes from and how crowdsourcing works.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|-------------------------------------|--|
| _____ 1. Bias | A. When an algorithm's output becomes its future input |
| _____ 2. Algorithmic bias | B. Raising money online from many people |
| _____ 3. Training data | C. Research done with help from non-scientists who contribute data |
| _____ 4. Machine learning | D. Data made freely available for anyone to use |
| _____ 5. Representative data | E. Bias that results from an algorithm's design or data |
| _____ 6. Proxy variable | F. Computers learning patterns from data instead of explicit rules |
| _____ 7. Feedback loop | G. The data a machine learning model learns from |
| _____ 8. Crowdsourcing | H. Getting input or information from many people via the Internet |
| _____ 9. Citizen science | I. Unfair preference for or against a group |
| _____ 10. Crowdfunding | J. Data that reflects the whole population it will be used on |
| _____ 11. Public data | K. A variable that indirectly stands in for another, like ZIP code for income |

Part 2 · Fill It In

Use each word from the word bank once.

data

bias

proxy

citizen science

crowdfunding

diverse

representative

crowdsourcing

1. Bias often enters an algorithm through its training _____.
2. Training data should be _____ of everyone who will use the system.
3. A _____ development team is more likely to notice bias.
4. Getting input from many people over the Internet is _____.
5. Volunteers counting birds for researchers is _____.
6. Raising money online from many small donors is _____.
7. Collecting more data from the same skewed source won't fix _____.
8. ZIP code can act as a _____ for income or race.

Name:

Class:

Date:

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Practice

Part 3 · Where Does the Bias Come From?

Check whether the bias mainly comes from the data or from the algorithm's design.

Item	Data	Algorithm design
1. A voice assistant trained mostly on one accent		
2. A programmer hard-codes “doctor” as male		
3. A photo app trained mostly on light-skinned faces		
4. A loan app rule rejects applicants from certain ZIP codes		
5. A health app’s “normal heart rate” built only from adult men		
6. A résumé filter programmed to reject any gap in employment		

Part 4 · Crowdsourcing or Not?

Example	Type
An online encyclopedia written by volunteers	_____
Thousands of people fund a new board game online	_____
Volunteers classify galaxy images for astronomers	_____
A company’s 5 engineers write an app	_____
Drivers report accidents in a traffic app	_____

Part 5 · Feedback Loop

A crime-prediction tool sends more police to neighborhoods with more past arrests. More police there leads to more arrests recorded there. Explain why this loop can make bias grow, and suggest one way to break it.

Part 6 · Design a Fair Project

You’re building a crowdsourced app to map broken sidewalks in your city. Describe how the digital divide could bias your data and two things you’d do to prevent it.



Think:

An algorithm trained on biased history will repeat that history — at computer speed and at massive scale.

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. What is crowdsourcing?
- A. Getting input or information from many people via the Internet
 - B. Encrypting data
 - C. Sorting a list
 - D. Storing data in the cloud

★★ On Level

2. A facial recognition system is much less accurate for some groups of people. What is the most likely cause?
- A. The users stood too far away
 - B. The cameras are broken
 - C. It used too much memory
 - D. Its training data didn't represent those groups well

★★ On Level

3. Which is an example of citizen science?
- A. Volunteers measuring rainfall in their backyards for climate researchers
 - B. A scientist working alone
 - C. A student watching a science video
 - D. A company hiring scientists

★★★ Exam Hard

4. An admissions algorithm doesn't use race, but it uses ZIP code. Why might it still be biased?
- A. Algorithms can't use numbers
 - B. ZIP codes can act as a proxy for race and income
 - C. ZIP codes change every year
 - D. ZIP codes are random

★★★ Exam Hard

5. Which TWO actions help reduce bias? (*Select two.*)
- A. Using data from only one city
 - B. Including people with diverse backgrounds on the team
 - C. Collecting more data from the same source
 - D. Testing with diverse users and comparing results

Part 8 · Written Response

Explain how biased data and the digital divide can combine to harm a group of people. Give a specific example.
